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MILITARY SPECIFICATION

PUNCHING MACHINE, METAL, TURRET TYPE, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC), metal punching machines of the turret type with equipment and accessories specified herein.

1.2 Classification. The machine covered by this specification shall be of the sizes listed in table I. The size required shall be as specified (see 6.2.1 and table I).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058- Machinery, Metal and Woodworking, Packaging of.

Benificaial comments (recommendations, additions, deletlons) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

CODE OF FEDERAL REGULATION (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATION SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards,

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001).

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which-are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

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AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One Inch Perforated Paper Tape.

RS-232 - Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interface.

RS-244 - Character Codes for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/ Positioning, Numerically Controlled Machines.

RS-358 - Subset of American National Standard Code for Information Interchanged for Numerical Machine Control Perforated Tape.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, D.C. 20006-1807.)

JOINT Industrial COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, Mclean, VA 22101-4269.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturer Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other information services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2,1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The punching machine shall be new (not a prototype) and one of the manufacturer's current production models. The punching machine shall be a CNC design which meets the requirements of this specification. The machine shall be an electrically powered, motor driven unit with a solid state soft-wired, integrated circuit, CNC system, The design shall include upper and lower multistation type turrets, mounted between the overhang and the base of a frame to form a workpiece opening between the turrets. The upper turret shall be provided with punch holders or lifter assemblies and the lower turret shall be provided with die holders. The machine shall be designed to allow the selected punch station to be rotated to a hole punching position under the machine drive ram. Turret rotation, workpiece positioning, and the punching action of the machine shall be controlled by the CNC system. The design shall provide convenient, efficient, and safe operation. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design, construction, and Measurement system. Unless otherwise specified (see 6.2.1), either the Inch-pound system or the International System of Units (SI) shall be used in the design and construction of the punching machine. In this specification, all dimensions, measurements, sizes, and capacities are given-in inch-pound units. These-dimensions may-be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

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3.2.1.1 Measuring and indicating device calibrations. Unless otherwise required, either the inch-pound or the SI system of measurement shall be used to 'graduate measuring and" indicating devices When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1).' Regardless-of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The punching machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the punching machine.

3.2.3 Energy efficiency. The punching machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the punching machine that present safety hazards, The safety devices shall not interfere with the operation of the punching machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the punching machine, whether or not specifically required herein, shall comply with all of the requirements of (OSHA) 29 CFR 1910 that are applicable to the machine itself. The punching machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial punching machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workplaces. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The punching machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the punching machine.

3.2.8 Environmental protection. The punching machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the punching machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be an oil level sight

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gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The punching machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the punching machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice,

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the punching machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the punching machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable “Detailed standard” section referenced therein.

3.3.7 Dials. All dials that indicate machine axes position shall be graduated in increments not larger than 0.001 inch, Dial circumferences shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 390.03 for the inch-pound system or ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper

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width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened steel. Hardness shall be not less than Rockwell C-48.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The punching machine shall consist of not less than a frame, table, machine drive, turret, punch holders, die holders, electrical system, motors and numerical-control-system.

3.4.1 Frame. The frame shall be constructed of either iron or steel and stress relieved. Unless otherwise specified, the frame shall be of the "C", "J", or bridge style having a base, column, and overhang to form a workpiece opening or throat. When only one style frame is acceptable, the particular style required shall be as specified (see 6.2.1). The frame shall be sufficiently rigid to withstand stresses imposed during performance of the punching requirements specified herein. The base of the frame shall have provisions for leveling and securing the machine to a flat mounting surface. When specified (see 6.2.1), the top of the frame shall be provided with a rigging eye or other means to facilitate lifting the machine.

3.4.2 Table. The machine shall be provided with a table for supporting the workpiece in the punching operation. The table shall be constructed of iron or steel with the top surface either hardened or provided with anti-friction bearings or rollers to facilitate workpiece travel across the table by the workpiece positioning unit. Table motion shall be accomplished on anti-friction bearings. The table motion and positioning shall be obtained by using anti-backlash ball screws and nuts, Table axial movement shall be accomplished by using servo drives with either an alternating current (AC) or direct current (DC) motors of the totally enclosed permanent magnet type. The servo motors shall have sufficient power and torque to position the table at the speed required in table I. The table shall be of the size required to fully support any workpiece size while located at any position within the range of the workpiece positioning unit. When required, the table top dimensions shall be of the size specified (see 6.2.1).

3.4.3 Machine drive. Unless otherwise required, the punching action of the machine shall be accomplished by listing either mechanical, hydraulic, or a combination mechanical and hydraulic type drive. When only one type drive is acceptable the particular type drive required shall be as specified (see 6.2.1).

3.4.3.1 Mechanical type drive. The mechanical type drive shall be provided with an electric motor driven assembly of mechanical components with connecting shafts, gears, chains, belts, or pulleys as required for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The components shall include a flywheel for supplying supplementary power to the drive motor. The flywheel shall be of cast iron or steel, and shall be dynamically balanced after machining or

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TABLE I - Sizes and capacity.

Capacity	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10	Size 11	Size 12	Size 13
Number of punch stations	20	10	20	22	20	24	33	33	16	42	16	60	36
Workpiece positioning unit range (without repositioning the workpiece within the workpiece positioning unit):													
In "X" axis, inches	40	39	40	40	40	48	48	60	30	51	80	74	80
In "Y" axis, inches	40	20	40	50	30	38	38	50	40	51	60	62	60
Workpiece positioning speed along the "X" and "Y" axes, inches per minute (IPM)	1970	787	2000	1970	2000	2000	3200	2000	2200	2500	2000	2500	2000
Hole punching capacity in SAE 1020 steel: Maximun diameter, inches, in material thickness, inches	3 .107	3.5 .075	3.5 .075	3.0 .134	3.5 .134	3.5 .134	3.5 .134	3.5 .134	3.5 .134	6 .107	4.0 .250	6.0 .134	4.5 .134
Diameter, inches in Maximum material thickness, inch	.8 .250	1.125 .250	2.0 .134	1.125 .250	1.165 .250	1.165 .250	1.165 .250	1.125 .375	1.0 .250	2.0 .250	1.0 .625	1.625 .375	1.750 .375
Hole per minute spaced on one inch centers in the maximum material thickness	200	120	200	200	180	200	280	225	180	240	95	200	200
Punched hole center line dimension accuracy inch	.005	.005	.005	.005	.005	.005	.005	.005	.004	.005	.005	.005	.005
Repeatability of hole location	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003
Workpiece size inches tonnage	40x40 16.5	20x39 20	40x40 20	40x48 22	40x30 30	48x38 30	48x38 30	60x50 30	30x40 34	49x49 33	80x60 45	72x60 45	80x60 50

NOTE: Machine size and capacities shall be not less than the stated requirements of the sizes ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the min- or less to the maximum or greater.

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precision machined concentric to their center. An eccentric or pitman shall be provided for converting the rotary motion of the motor and flywheel to the reciprocating motion required by the ram, All rotating and sliding components shall be provided with bearings or bushings with means for positive and continuous lubrication. The energy of the drive shall be controlled by a clutch and brake arrangement that is capable of transmitting the maximum developed torque to the ram, Provisions shall be provided to automatically control the ram stroke. The ability to control the cycle by reversing under pressure shall also be provided.

3.4.3.2 Hydraulic type drive. The hydraulic type drive shall be provided with a hydraulic power unit for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The hydraulic power unit shall include an electric motor, hydraulic pump, fluid reservoir, ram cylinder unit, and connecting hose or tubing. The fluid reservoir shall be provided with a fluid level gage, and a hydraulic fluid filter with replaceable type element. The hydraulic cylinder unit shall be the double acting type. The drive shall be provided with hydraulic fluid control devices to include a high pressure relief valve. Connecting hoses or tubing shall be capable of withstanding fluid pressure up to 150 percent of the rated working pressure of the power unit. The system shall conform to the requirements of JIC H-1 Hydraulic Standards.

3.4.4 Turrets. The turrets shall be stress-relieved castings of iron or steel, mounted on anti-friction type bearings, and directly supported by the machine frame. The turrets shall be synchronized to rotate as a single unit with the punch stations of the upper turret constantly aligned with the corresponding die stations of the lower turret. Turret rotation shall be bi-directional and accomplished by a power unit other than that provided for the machine drive. The machine shall be provided with indexing pins for aligning and locking the turrets in the punching position. The mating holes of the indexing pins shall be provided with replaceable bushing inserts. When the turrets are locked in the punching position, the punch holders of the upper stations shall be concentrically aligned with the die holders of the corresponding lower die stations to within 0.003 inch.

3.4.5 Punch holders. Each punch station of the upper turret shall be furnished with a punch holder. The punch holders shall be steel, designed for accurate seating in the upper turret to assure conformance with the alignment requirements of 3.4.4. Means shall be provided for radially locating and securing the punches in the punch holder. The punch holders shall be designed to accommodate punches for punching round, square, and rectangular shaped holes.

3.4.6 Die holders. Each die station of the lower turret shall be furnished with a die holder or other suitable means for holding the dies in position. The die holders shall be either steel or cast iron, designed for accurate seating in. the lower turret to assure conformance with the alignment requirements of 3.4.4. Means shall be provided to radially locate and lock the dies and the die holders. The die holders shall accommodate dies for punching round, square, and rectangular shaped holes.

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3.4.7 Workpiece strippers. Machines that are not provided with self-stripping type punches shall be provided with strippers that prevent the workpiece from riding-up with the punches when they are withdrawn from the dies.

3.4.8 Punches and dies. When required, Punches and dies of the sizes and shapes specified (see 6.2.1), shall ‘be furnished. The punches and dies shall be tool steel, hardened and-ground for accuracy. The punches and dies shall be permanently marked for size and identification. “

3.4.9 Cutting Unit. When required the machine shall be furnished with either a plasma arc or carbon dioxide (CO^2) laser cutting unit. The cutting unit requirements shall be fully described as required (see 6.2.1), The plasma arc or CO^2 laser cutting head shall be mounted between the overhang and base of the frame to form a workpiece opening between the turrets.

3.4.10 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.10.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase 60 Hz circuit, Unless otherwise specified (see 6.2.1), the machine shall be initially wired to operate from a 460 volt source.

3.4.10.2 Motors. Motors shall be rated for continuous duty and shall have . ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements for a dripproof enclosure. Alternating current (AC) motors shall be designed to operate on 60 hertz (HZ). All motors shall meet the requirements of NEMA MG-1.

3.4.10.3 Solid State components. Solid state design shall be used throughout for electronic components.

3.4.10.4 Control circuit voltage. Control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.4.10.5 Hour meter. Unless otherwise specified (see 6.2.1), the punching machine shall be equipped with an hour meter. The hour meter shall be installed to display ’accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the punching machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

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3.4.11 Numerical control system, The numerical control system shall be of the solid state, soft-wired, micro processor, computerized numerical control type. The system shall provide automatic control of machine functions, operating modes, punch selection, and two axes control of the horizontal "X" and "Y" motion, When a plasma - arc or carbon dioxide laser cutting head is furnished, the cutting head shall also be controlled by the CNC unit, The axes shall be identified in accordance with EIA standard RS-267. The control shall be a positioning and contouring type having linear and circular interpolation, capable of simultaneous two axis contouring. Unless executive program routines are provided in nonvolatile memory form (i. e. bubble memory cassette, hard disc, etc), the control unit shall be provided with absolute and incremental programming. Control features shall include buffer storage, part program storage, program edit capability and control diagnostics. The control unit shall direct machine functions from part program data in memory storage, punched tape, and manual data input. The control shall initiate a halt or an error signal should a fault condition occur in the control, All necessary executive program routines shall be furnished in the form of 1 inch, eight channel, binary coded decimal punched tape for controls having volatile memory. A line voltage variation of + 10 percent from normal shall not adversely affect the control system functins, The control shall function in ambient temperatures ranging from 50 degrees F to 110 degrees F and relative humidities ranging from 5 percent to 95 percent. The control shall automatically shut down when internal operating temperatures exceed safe operating temperature. Control resolution shall be not greater than 0.001 inch for each linear axis. The auxiliary functions of the system shall include, not less than the punch action and opening and closing of the workpiece clamps. The system shall be provided with operational interlocks and functional protection to prevent the machine from operating when the turrets are not locked in the punching position, the machine drive brake is not released, or the air pressure of the machine or auxiliary drive units drops below the required operating level. Interlock circuitry shall also be provided for assuring sequential operation of the system, The control unit shall rotate the turrets, lock the programmed punch and die station in the punching position, and position the workpiece throughout the full range of travel along the "X" and "Y" axes of motion. The control unit shall automatically control the acceleration and deceleration of all axes of motion. The control unit shall control axis movements individually, simultaneously, or in any combination. The control unit shall control the positioning units for accomplishing any function within the capacity of the machine without loss of cycle synchronization. The smallest program increment for positioning the workpiece shall be not more than 0.001 inch and the largest shall be not less than the workpiece positioning range specified in table I.

3.4.11.1 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232.

3.4.11.2 Data input. Unless otherwise specified (see 6.2.1), the data input media shall be one inch, eight channel, punched tape in conformance with EIA Standards RS-227, RS-244, RS-274 and RS-358.

3.4.11.2.1 Tape reader. A photoelectric tape reader shall be provided. The tape reader shall feed, and read tape prepared in accordance with paragraph 3.4.11.1, at a rate of not less than 150 characters per second. The tape reader shall be housed in a dust-free area of the machine control

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unit and shall be easily accessible for maintenance and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display unit to indicate fault condition.

3.4.11.2.2 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.11.3 Control panel. The control unit shall have an operator control panel for displaying the control functions and visual signals of the numerical control system and the machine tool. The control panel shall be provided with knobs, switches, or digital input for controlling system functions and shall have not less than power-on and off, cycle start-stop, emergency stop, optional stop, mode selection, feed hold, feedrate override, jog, jog direction, and end of block. Indicator lights or the data display unit shall indicate or display not less than reading error, in-position, over-temperature, tool change, program stop, end of program, turrets not locked in punching position, and low air or hydraulic pressure if applicable.

3.4.11.4 Data display. Either a cathode ray tube (CRT) or a universal alphanumeric data display unit shall be provided. The CRT unit shall display not less than 250 characters simultaneously. The universal alphanumeric display unit shall be capable of being used in a variable scanning mode to interrogate all registers sequentially or held to display only a particular register as the content changes. Either unit shall display, as a minimum, the following data:

a. Sequence number readout.
b. Actual position of each axis. The readout shall at least display the accumulative command axis position of one axis at a time without loss of reference dimensions as a result of the switching.
c. Control and system error message.
d. Miscellaneous functions.

3.4.11.5 Control system modes of operation. The operational modes of the control unit shall include manual, automatic, single block, MDI, and manual jog.

3.4.11.6 Absolute/ incremental input. The control shall have absolute/incremental input capability, This mode shall be selected by a preparatory "G" code which may be switched during the tape program,

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3.4.11.7 Automatic tape code recognition. The control shall automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes.

3.4.11.8 Switchable inch/metric input. The control system shall provide switchable inch-metric Input capability.

3.4.11.9 Part program storage. The machine control unit shall be capable of storing in memory the content of not less than 150 feet of postprocessed part program tape data.

3.4.11.10 Standby battery power. Unless nonvolatile part program memory is provided, standby battery power shall be provided to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.11.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion, or modification of any part program in the part program memory data related to part program. Edit shall be input by the MDI provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.11.12 Contouring feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 100 percent of-the programmed rate in either continuous adjustment or in steps of not more than 25 percent. The feed rate override control shall have a position that allows return of feed rate to that programed.

3.4.11.13 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.11.14 Mirror image. When specified (see 6.2.1), the machine shall be capable of producing either right or left-hand parts from the same tape data through the use of inversion switches for each axis. Inversion of each axis may be by keyboard entry.

3.4.11.15 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.11.16 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.11.17 Block delete. The control shall have a block delete feature to permit by-passing blocks of programmed data.

3.4.11.18 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

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3.4.11.19 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.11.20 Lead and reversal error compensation. Lead error and reversal error compensation shall be provided to permit data to be stored in the control to compensate for errors and backlash of each linear axis.

3.4.11.21 Additional features. The following additional control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Leading zero suppression or decimal point programming.
d. Inches per minute (IPM)/millimeters per minute (MMPM), programing in all modes of operation.
e. Pattern repeat.

3.4.11.22 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.11.23 Set zero. The control unit shall have a “set zero” feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.11.24 Tape tryout. The control shall have a tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.11.25 Operational software. Operational software shall be furnished with the control and shall contain all logic necessary to effectively operate the machine tool. The software shall be provided either resident in the control in the form of nonvolatile memory or separately in the form of one inch, eight channel, binary coded decimal punched tape for controls having volatile memory. In either case, the software shall be provided for meaningful diagnostic relating to the use of the controls, machine tool part program and operator’s actions.

3.4.11.26 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures to not less than board level.

3.4.11.27 Positioning units. The primary command functions of the system shall be accomplished by turret and workpiece positioning units with positioning ranges and speeds to comply with table I. Positioning unit travel shall be obtained by electric motor driven anti-backlash ballscrews and nuts. Response of the positioning units to the primary commands for starting and stopping and the rate of motion shall be such that the specified punching accuracy is maintained. Except for sizes 1 and 3, which may be unidirectional, the turret positioning unit shall be provided with hi-directional turret rotation capability for automatically rotating the turrets in the direction for which the distance is shortest between the programmed punch and die station and the punching position under the ram. The workpiece

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positioning unit shall hold the workpiece by at least two clamps and shall be adjustable for holding any workpiece size ranging up to the capacity of the machine. The workpiece positioning unit shall be provided with means to prevent the workpiece clamps from being punched during the punching operation. When specified (see 6.2.1), the workpiece positioning unit shall be provided with automatic sheet repositioned characteristics for location of sheets longer than the rated capacity of table I by tape command. The closed loop workpiece positioning unit shall measure and feed back to the control unit, the output signals of actual workpiece position to be compared with the input positioning commands of the program tape data. The control unit shall be arranged to shut down the machine for positioning errors in excess of 0.003 inch.

3.4.11.28 Postprocessor. When specified (see 6.2.1), a postprocessor shall be furnished with the capability of converting the coordinate data generated by the computer program into machine language format of the punching machine. The computer and operating software, on which the postprocessed is to operate, shall be fully described under 3.9.

3.4.11.29 Test tape. Unless otherwise specified (see 6.2.1), a test tape and print out, programmed to perform all system functions specified herein shall be furnished. The test tape shall operate all slides, motors, pumps, adjusting mechanisms, and accessories during each cycle of the test tape.

3.5 Sizes and capacity. Unless otherwise specified (see 6.2.1), the size and capacity shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of automatic punching holes in SAE 1020-steel to meet the capacity and accuracy requirements of table I as applicable for the machine size specified. The numerical control system shall meet the performance requirements of 3.4.11,

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in 3.4.4.

3.8 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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3.11.1 Lubrication plate. A corrosion resistance lubrication plate shall be securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
viscosity

3.11.2 Nameplate. A corrosion resistance nameplate shall be securely attached to each machine, The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the punching machine and accessories shall meet all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. he inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the qualiity conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The punching machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than one hour, during which time the machine drive, and turrets shall be actuated and observed to determine compliance with the requirements specified herein. The CNC system shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. Run-off checks of all mechanical functions of the machine and CNC system functions shall be made in each mode of operation specified herein. Upon completion of the warm-up check procedure, the system shall be cycled continuously for a period of not less than two hours under tape control while programmed to demonstrate reliability of the equipment. There shall be no failure or malfunction of the machine or control. Should a failure or malfunction occur, the test shall be started from setpoint and retimed from zero hours. The program tape used for cycling shall position the machine over the entire working range and shall include operation of all controlled functions of the machine. The restart of each cycle shall occur within one minute of completion. At the end of the two hour test period, the machine shall return to setpoint within an accuracy of ± 0.005 inch. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period. All tapes used for the test shall become the property of the Government.

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4.7.1.1 Punch and die station alignment. Each punch and corresponding die station of the turrets shall be positioned and locked in the punching position. Turret position shall meet the turret alignment requirements of 3,4.4.

4.7.2 Machine performance test. The machine shall be tested as follows to determine compliance with the performance requirements of table I, as applicable for the size machine specified. When hole punching is required, SAE 1020 steel shall be used.

4.7.2.1 Positioning speed. Positioning speed shall be verified by clamping the maximum capacity size workpiece in the workpiece positioning unit through the full positioning range for the "X" and "Y" axes. The time required and distance traveled shall conform to table I.

4.7.2.2 Hole punching capacity. Hole punching capacity shall be verified by operating the machine for punching at least 12 holes of each of the diameters specified for each of the material thicknesses specified in table I.

4.7.2.3 Hole punching speed and accuracy. Hole punching speed and accuracy shall be verified by operating the machine to continuously punch 0.500 inch diameter holes, spaced on one inch centers in a workpiece with a thickness of not less than the maximum material thickness capacity of the applicable size machine. The holes shall be punched for a period of one minute. Upon completion, the number of holes punched shall be counted and the resulting punching accuracy shall be measured to determine compliance with the specified requirements.

4.7.2.4 Repeatability test. Two workplaces of 1/8 inch thick material shall be punched in consecutive order with 24 holes, each using a program tape with a hole pattern of the supplier's choice. The two plates shall be inspected to verify the repeatability requirements of table I.

4.7.3 CNC performance test. The system shall be tested to determine compliance with the performance requirements of 3.6. The control unit shall be actuated to operate the machine under the automatic, semi- automatic, and manual modes.

4.7.3.1 Automatic. The test tape specified in 3.4.11.28 shall be inserted in the control unit and the system operated to perform the functions specified. The print out furnished shall be used to verify the required functions.

4.7.3.2 Semi-automatic. The control unit shall be operated as specified in 4.7.3.1, but with at least six separate insertions of commands that require the machine to perform positioning functions. The print out furnished shall be used to verify the functions required.

4.7.3.3 Manual. The control unit shall be manually operated to verify each of the primary and auxiliary functions of the system.

4.7.4 Postprocessor validation. When a postprocessor is furnished (see 3.4.11.27), the postprocessor shall be validated by the supplier to assure that it is compatible with the computer configuration specified to produce operating programs (tape) that are acceptable to the numerical control

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system/machine combination for accomplishing all programmable functions for which they are designed. All postprocessor and test program errors shall be corrected by the supplier,

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for punching holes in metal using automatic control for hole punching workpiece and turret positioning. The machines may be used for repetitive high production punching but are best suited for prototype, short run punching.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2 and table I).

c. First article, if required (see 3.1)

d. If machine design and construction is required in a specific dimensional system (inch-pound or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch-pound or SI) state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g" Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9)

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i. If machine frame style is to be a specific type, state required style (see 3.4.1).

j. Rigging eye, if required (see 3.4.1).

k. Table top dimensions, if required (see 3.4.2).

1. If machine is required to have a specific drive, mechanical or hydraulic, state required type (see 3.4.3).

m. Punches and dies, specify shapes and sizes required (see 3.4.8).

n. Cutting unit, if required, state required type (see 3.4.9)

o. Input power supply, if different (see 3.4.10.1).

p. Motor enclosure, if different (see 3.4.10.2).

q. Hour meter, if different (see 3.4.10.5)

r. Peripheral equipment interface, if different (see (3.4.11.1)

s. Data input, if different (see 3.4.11.2).

t. Mirror image, if required (see 3.4.11.14).

u. Automatic sheet repositioned, if required (see 3.4.11.27).

v. Specify and fully describe the postprocessor, if required (see 3.4.11.28).

w. Test tape, if different (see 3.4.11.29).

x. Size and capacity, if different (see 3.5).

y. Additional performance features, if required (see 3.6.1).

z. Standard equipment, if different (see 3.8).

aa, Specify and fully describe optional equipment, if required (see 3.9).

bb . Specify applicable repair parts, if required (see 3.10).

cc. First article inspection, if different (see 4.4).

dd. Quality conformance inspection, if different (see 4.5).

ee. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

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6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be-specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. Specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article shall be provided in the contract.

6.4 Cross reference of classification changes

MIL-P-80054C	MIL-P-80054D
Size 1	Size 1
	Size 2 (New size)
Size 2	Size 3
	Size 4 (New size)
Size 3	Size 5
Size 4 (Deleted)	Size 6 (New size)
Size 5 (Deleted)	
Size 6 (Deleted)	
Size 7	Size 7
Size 8	Size 8
Size 9 (Deleted)	Size 9 (New size)
Size 10 (Deleted)	
Size 11	Size 10
	Size 11 (New size)
Size 12 (Deleted)	Size 12 (New size)
Size 13	Size 13

6.5 Subject term (keyword) listing.

Data display
Hydraulic drive
Mechanical drive
Numerical controls
Punching machine, metal
Standby battery power
Tape reader
Turret type
Workpiece strippers

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3445-0109)

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Review activity:
Air Force - 84-71

User activity:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80054D	Punching Machine, Metal, Turret Type, Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation.

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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